Pneumonia Diagnosis Detection and Localization

Using a Custom CNN and Grad-CAM

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1. Problem Statement

Chest X-ray images can contain opacity regions that are difficult to identify consistently. This project develops a machine learning pipeline that classifies chest X-rays into Normal and Lung Opacity classes and provides weakly supervised localization of the regions contributing to the Lung Opacity prediction. The approach uses a custom convolutional neural network (CNN) trained from scratch, followed by Grad-CAM to generate class-discriminative localization maps. Predicted regions are converted into bounding boxes and compared with RSNA ground-truth annotations using Intersection over Union (IoU).

2. Dataset

The project uses the RSNA Pneumonia Detection Challenge dataset. For this one-week implementation, a balanced subset of 2,000 images was selected after filtering the label information to retain the Normal and Lung Opacity classes and removing duplicate patient IDs. The selected data contain 1,000 Normal images and 1,000 Lung Opacity images.

Split	Images
Training	1,400
Validation	300
Test	300

3. Approach and Methodology

The complete pipeline was implemented in PyTorch and executed in a GPU-enabled Kaggle environment. The main stages are:

- Preprocessing: DICOM chest X-rays were intensity-normalized, converted to RGB, resized to 224 x 224 pixels, and normalized using mean and standard deviation of 0.5. Training images used random horizontal flips and small rotations.
- Custom CNN: Four convolutional layers with ReLU and max-pooling were followed by global average pooling, dropout, and a two-class fully connected output layer. No pretrained model, ImageNet weights, or transfer learning was used.
- Training: The network was trained for 5 epochs with cross-entropy loss, Adam optimizer, learning rate 0.001, and batch size 32. Validation data were used to monitor performance during training.
- Localization: Grad-CAM was applied to the final convolutional layer (conv4). The activation map was thresholded to obtain a predicted bounding box for the Lung Opacity region.
- Evaluation: Classification was measured using accuracy, precision, recall, F1-score, and ROC-AUC. Localization was assessed using IoU between the predicted box and the ground-truth region.

4. Implementation Overview

The notebook is organized as a reproducible end-to-end workflow: dataset loading and filtering, patient-level deduplication and stratified splitting, preprocessing and DataLoaders, custom CNN definition, training and validation, test-set evaluation, Grad-CAM generation, bounding-box construction, IoU evaluation, and threshold analysis. The final model weights and result summaries are also saved from the notebook for reuse and submission packaging.

5. Results

The custom CNN was evaluated on the unseen 300-image test set. The resulting classification metrics are:

Metric	Result
Accuracy	78.67%
Precision	76.88%
Recall	82.00%
F1 Score	79.35%
ROC-AUC	82.07%

The test-set confusion matrix was [[113, 37], [27, 123]], corresponding to 113 correctly classified Normal cases and 123 correctly classified Lung Opacity cases.

6. Grad-CAM Localization and IoU

Grad-CAM was used to visualize the image regions that most influenced Lung Opacity predictions. The resulting activation maps were converted into bounding boxes through thresholding and contour-based region extraction. The predicted boxes were compared with the ground-truth opacity annotations using IoU. In the localization experiment, 20 Lung Opacity test cases were inspected; 16 were classified as Lung Opacity and were therefore included in the positive localization analysis.

7. Threshold Experiment

Five Grad-CAM thresholds (0.3, 0.4, 0.5, 0.6, and 0.7) were evaluated on the same localization subset. The best threshold was selected using mean IoU.

Threshold	Cases	Mean IoU	Median IoU	Best IoU	Worst IoU
0.3	16	0.1877	0.1367	0.4744	0.0142
0.4	16	0.1888	0.1218	0.4865	0.0178
0.5	16	0.1743	0.0927	0.4869	0.0000
0.6	16	0.1510	0.0825	0.4704	0.0000
0.7	16	0.1543	0.0464	0.6408	0.0000

The selected threshold is 0.4, which achieved the highest mean IoU of approximately 0.1888 among the evaluated thresholds. The localization experiment is intentionally described as preliminary because it uses a limited subset of cases.

8. Conclusion

A custom CNN was successfully developed and trained from scratch for binary classification of chest X-ray images into Normal and Lung Opacity classes. The model achieved 78.67% test accuracy and 82.07% ROC-AUC. Grad-CAM provided weakly supervised localization without requiring a separate object-detection model, and the threshold study identified 0.4 as the best setting among the tested values. The results demonstrate an end-to-end classification plus localization workflow while also showing that localization quality remains an area for improvement.

9. Limitations and Future Work

- The implementation uses a subset of 2,000 images rather than the complete RSNA dataset.
- Localization was evaluated on a limited set of 20 Lung Opacity test cases, with 16 positive predictions included in the summarized positive localization analysis.
- Future work can scale training to the full dataset, increase localization evaluation coverage, improve the custom CNN architecture, and explore more robust localization and post-processing methods.